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AIUB, Physics 2 Mid term

#Physics Two

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Temperature, heat and the first law of thermodynamics.

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Thermodynamics :-

⇒ One of the principal branches of physics and engineering is thermodynamics. which is the study and application of the thermal energy of systems. One of the central concepts of thermodynamics is temperature.

Temperature :-

⇒ Temperature is an SI base quantity related to our sense of hot and cold.

We use thermometer to mesure temperature, which has special substance inside that changes in predictable way as its gets hotter or colder.

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Relationship between Temperature Scales

$$\boxed{\frac{C}{5} = \frac{F-32}{9} = \frac{K-273}{5}}$$

$$T_C = T - 273.15^\circ$$
$$T_F = \frac{9}{5} T_C + 32^\circ$$

Where:-

T = Temperature in kelvin scale

T_C = Temperature in Celsius scale

T_F = Temperature in Fahrenheit scale.

Triple point of water:-

$K = 273.16^\circ K$

$C = 0.01^\circ C$

$F = 32.02^\circ F$

Absolute Zero:-

$K = 0° K$

$C = -273.15° C$

$F = -459.67° F$

Q:- The temperature of an object is 25°C. After absorbing some amount of heat the temperature of an object becomes 55°C. What would be the temperature difference in kelvin scale, in this case?

⇒ we know:-

$T_C = T_1 - 273$

$T_1 = T_C + 273$

$= 25 + 273$

$= 298 K$

Again:-

$T_C = T_2 - 273$

$T_2 = 273 + 55$

$= 328$

The temperature difference $= (328 - 298) K$

$= 30 K$

Ans:-

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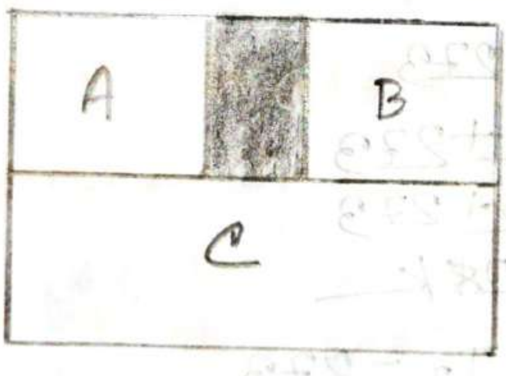
The Zeroth Law of Thermodynamics:—

⇒ The Zeroth Law of thermodynamics says that if two things are at the same temperature as a third things, then they are also at the same temperature as each other. It helps us understand how hot or cold things are and compare there temperatures.



$T_A = T_C$ and $T_B = T_C$

so, $T_A = T_B$

Note:— Thermometers are made using this formula.

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#ABsorption of Heat:—

Temperature and Heat:—

⇒ A change in temperature is due to a change in the thermal energy of the system because of a transfer of energy between the system and the system's environment.

The transferred energy is called 'Heat' and is symbolized Q.

'Heat is positive':—

⇒ when energy is transferred to a system's thermal energy from its environment. that means Heat absorbed by the system.

'Heat is negative':—

⇒ When energy is transferred from a system's thermal energy to its environment. that means Heat is released by the system.

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Heat:-

⇒ Heat is the energy that moves from somthing hot to somthing cooler because they have diffrent temperatures.

$$1\ cal = 3.968\times10^{-3}\ Btu = 4.1868\ J$$

The Absorption of Heat by Solids and Liquids

Heat Capacity:-

⇒ The Heat capacity, C of an object is the proportionality constant between the heat Q that the object absorbs or loses and the resulting temperature change ΔT of the object. that is

$$Q \propto \Delta T$$

$$Q = C\Delta T$$

$$\therefore \boxed{C = \frac{Q}{T_f - T_i} = \frac{Q}{\Delta T}}$$

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Where,

C = heat capacity

Q = heat.

SI Unit = cal/C° or J/K

Specific Heat Capacity:-

⇒ Heat capacities of objects made of the same meterial are proportional to there masses. We use specific heat capacity per unit mass (c) to compare diffrent amounts of the material. [$c = C/m$ or $C = mc$]

We know:-

$$Q = C\Delta T$$

$$Q = mc \times \Delta T \quad [C = mc]$$

$$\boxed{c = \frac{Q}{m \times \Delta T}}$$

$$c = \frac{dQ}{m\,dT}$$

Where:-

c = Specific Heat capacity

Q = Heat

m = mass

C = total heat capacity.

SI Unit = cal/g-c° or J/Kg-K

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Molar specific Heat :-

⇒ The molar specific heat of a material is the heat capacity per mole, which means per 6.02×10^{23} elementary units of the material.
When quantities are expressed in moles, specific heats must also involve moles they are then called molar specific heats. and is Symbolized c_m.

$$\therefore Q = n c_m \Delta T$$

$$\boxed{c_m = \frac{Q}{n \Delta T} = \frac{dQ}{n\,dT}}$$

where:-

c_m = molar specific Heat
Q = Heat
ΔT = Temperature change
n = number of moles

SI Unit :-

cal/mol-C° or J/mol-K

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Problem 23:—

⇒ A small electric immersion heater is used to heat 100g of water for a cup of instant coffee. The heater is labeled 200 watts. Calculate the time required to bring all this water from 23°C to 100°C, ignoring any heat losses.

⇒ is given:—

$m = 100g = 100\times10^{-3}$ kg

$T_f = 100+273$, $T_i = 23+273$

$P = 200\,W$

We know:—

Specific heat capacity of water $C = 4200$ J/kg-K

Now, we know:—

$C = \frac{dQ}{m\,dT}$

$Q = C\times m\times(T_f - T_i)$

$= 4200\times100\times10^{-3}\times(373-296)$

$= 420\times77$

$= 32340$ J

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Again We know:—

$$p = \frac{Q}{t}$$

$$\therefore t = \frac{Q}{P}$$

$$= \frac{32340}{200}$$

$$= 161.7 \text{ Sec.}$$

Ans:—

Problem 24:—

⇒ A certain Substance has a mass per mole of 50 g/mol. When 314 J is added as heat to a 30 g sample, the sample temperature rises from 25°C to 45°C. What are the

(a) specific heat?

(b) molar specific heat of this substance?

(c) How many moles are in the sample?

⇒ in given:—

mass of a sample $m = 30 \times 10^{-3}$ kg

mass per Mole $M = 50 \times 10^{-3}$ kg/mol

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$T_f = 45 + 273K = 318K$

$T_i = 25 + 273K = 298K$

$\Delta T = T_f - T_i = 20K$

$Q = 314 J$

(a)

we know,

$$C = \frac{Q}{m\Delta T}$$

$$= \frac{314}{30\times10^{-3}\times(20)}$$

$$= 523.33 \ J/kg\text{-}K$$

(c)

we know,

number of mole $n = \frac{m}{M}$

$$= \frac{30\times10^{-3}}{50\times10^{-3}}$$

$$= 0.6 \ mol$$

(b)

we know,

$$C_m = \frac{Q}{n\Delta T} = \frac{314}{0.6\times20} = 26.167 \ J/mol\text{-}K$$

Ans:

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Heats of transformation:—

Specific Heat:—

⇒ Heat required to change the temperature of the material or sample.

$$Q = cm\Delta T$$

$$c = \frac{Q}{m\Delta T}$$

Laten Heat: সুপ্ত তাপ:—

⇒ Heat required to change the state of material.

Laten Heat of Fusion:—

Solid → Liquid

Liquid → Solid

Laten Heat of Vaporization:—

Liquid → Vapor

Vapor → Liquid

$$\boxed{Q = mL}$$

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Heat of Vaporization:—

⇒ The Heat of Vaporization L_v is the amount of energy per unit mass that must be added to Vaporize a Liquid or that must be removed to condense a gas.

$$\therefore \boxed{Q = mL_v}$$

where—

L_v = The heat of Vaporization

m = mass

$$\boxed{L_v = 2.26 \times 10^6 \ J\,kg^{-1} = 580\ cal/g = 40.7\ kJ\,mol^{-1}}$$

$L_v = 2256 \times 10^3\ J\,kg^{-1}$

Heat of Fusion:—

⇒ The Heat of fusion L_f is the amount of energy per unit mass that must be added to melt a solid or that must be removed to freeze a liquid.

$$\therefore \boxed{Q = mL_f}$$

$$\boxed{L_f = 3.36 \times 10^5\ J\,kg^{-1} = 79.5\ cal/g = 6.01\ kJ/mol}$$

$L_f = 333 \times 10^3\ J\,kg^{-1}$

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Problem 278—

⇒ Calculate the minimum amount of energy, in joules, requried to comp letely melt 130g of silver initially at 15°C. Consider melting temperature of silver is 1235K and its specific heat capacity and latent heat of fusion are 236 J/kg-K and 105 kJ/kJ, respectively.

⇒ is given:—

$$T_i = 15+273 = 288K$$

$$T_f = 1235\ K$$

$$\Delta T = T_f - T_i$$

$$= 1235 - 288$$

$$= 947K$$

$$m = 130g = 130\times10^{-3} kg$$

$$c = 236\ J/kg\text{-}K$$

$$L_f = 105 kJ/kJ = 105\times10^{3}\ J/kg$$

We know:—

$$c = \frac{Q_1}{m\Delta T}$$

$$c = \frac{Q_1}{m\Delta T}$$

$$\therefore Q_1 = c\times m\times\Delta T$$

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$= 236 \times 130 \times 10^{-3} \times 0.47$

$= 22.05 \times 10^{3}$ J

Again, we know,—

Latent heat of fusion $Q_2 = mL_f$

$= 130 \times 10^{-3} \times 105 \times 10^{3}$

$= 13.65 \times 10^{3}$ J

The total heat requried.

$Q = Q_1 + Q_2$

$= (22.05 \times 10^{3} + 1.36 \times 10^{4})$ J

$= 42700$

$= 42.7 \times 10^{3}$ J

Ans:—

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Problem 28:—

⇒ How much water remains unfrozen after 50.2 kJ is transferred as heat from 260g of liquid water initially at its freezing point?

⇒ in given:—

$Q = 50.2 \times 10^3 J$

Liquid water, $m = 260 \times 10^{-3} kg$ & $L_f = 3.36 \times 10^5 J kg^{-1}$
$= 333 \times 10^3 J kg^{-1}$

We know:—

$Q = m' L_f$

$m' = \frac{Q}{L_f}$ [m' = mass of frozen water]

$= \frac{50.2 \times 10^3}{333 \times 10^3}$

$= 0.15 kg$

The amount of water that remains unforzen $= m - m'$

$= (260 \times 10^{-3} - 0.15) kg$

$= 0.11 kg$

$= 110 g.$

Ans.

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Thermodynamic proceess.

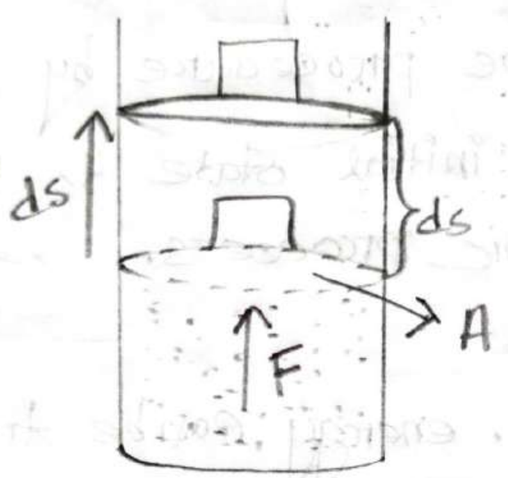
⇒ If a system starts from an initial state i, described by a pressure P_i, a Volume V_i and a Temperature T_i. Now we want to change the system to a final state f, described by a pressure P_f, a Volume V_f, and a Temperature T_f. The procedure by which we change the system from its initial state to its final state is called a thermodynamic proceess.

* During a process, energy can be transferred into the system from a heat source (positive heat) or transferred out of the system (negative heat)

* Also work can be done by the system to raise the loaded piston (positive work) or lower it (negative work).

Work done in Compression and expansion :-

Work is done by the system on the environment :-



We know :-

$$\text{pressure } p = \frac{F}{A}$$

$$\text{so, } F = PA$$

$$\therefore dW = \vec{F} \cdot \vec{ds}$$
$$= F \cdot ds \cdot \cos 0°$$
$$= PA \cdot ds$$

$$\boxed{dW = P \cdot dV} \qquad [\, dV = A \cdot ds \,]$$

Where, F = Force
P = pressure
A = Area of the piston
dv = differential change in the Volume.

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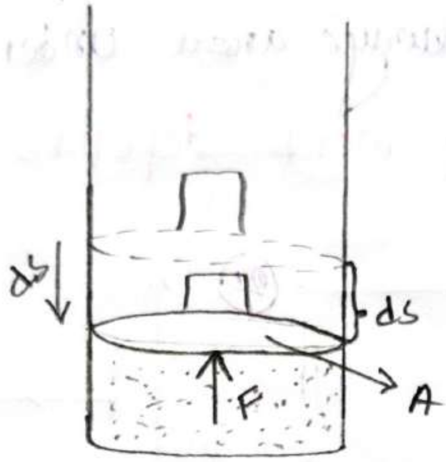
☒ Work is done on the system by the envircoment :—



We know :— $P = \frac{F}{A}$

$F = PA$

$\therefore \quad dW = \vec{F} \cdot \vec{ds}$

$= F \cdot ds \cdot \cos 180^\circ$

$= - F \cdot ds$

$= - PA \cdot ds$

$$\boxed{dW = - P \cdot dV}$$

☒ the total work done by the any gas is :—

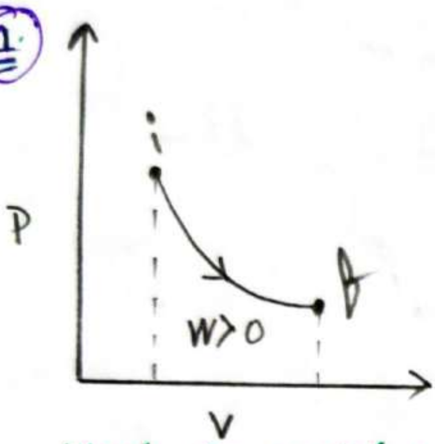
$$\boxed{W = \int dW = \int_{V_i}^{V_f} P \cdot dV}$$

Path dependent work done in terms of p-V diagram:—

⇒ Work done is always area under the p-V curve:—

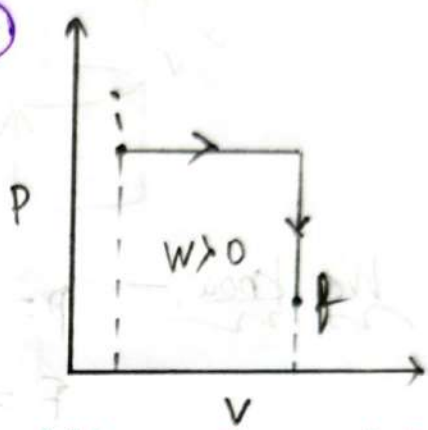
→ The amount of work depends on path.

1.



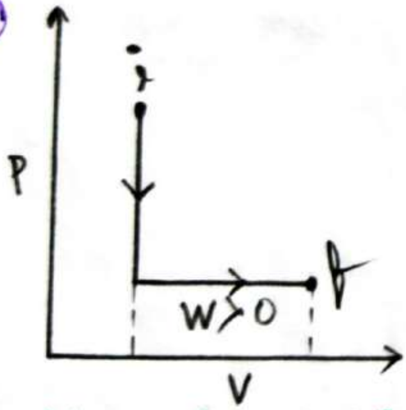
Work is positive.

2.



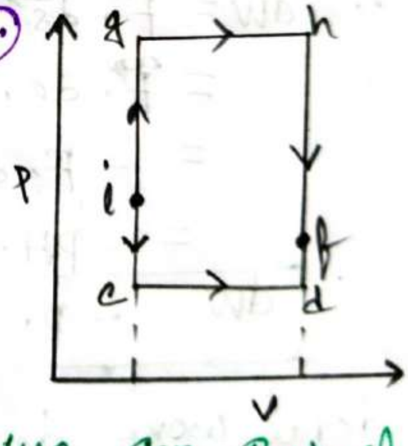
Work is positive.

3.



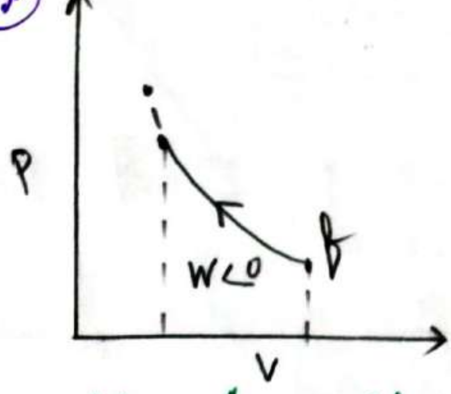
Work is positive

4.



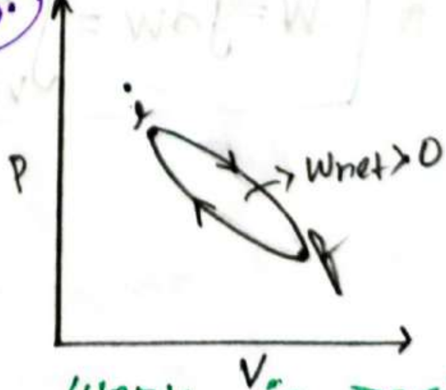
We can control how much work it does.

5.



Work is negative.

6.



Work is positive cz cycling clockwise

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Calculating work done from pv Diagram:-

1. Always, work done equals the area under the pv curve.

2. At constant volume, $\Delta V = 0$, $W = 0$

3. At constant pressure $W = p\Delta V$

4. When both volume and pressure changes, we have to consider the area concept only.

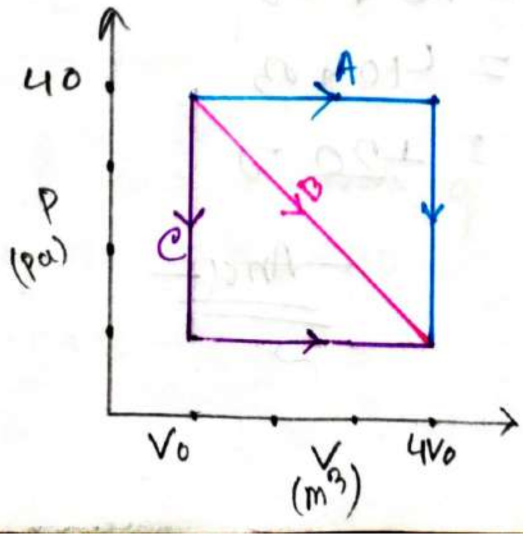
Problem: 43:-

in Fig., a gas sample expands from V_0 to $4V_0$. while its pressure decreases from P_0 to $P_0/4$. If $V_0 = 1m^3$ and $P_0 = 40$ pa how much work is done by the gas if its pressure changes with volume via

(a) Path A.

(b) Path B.

(c) Path C.



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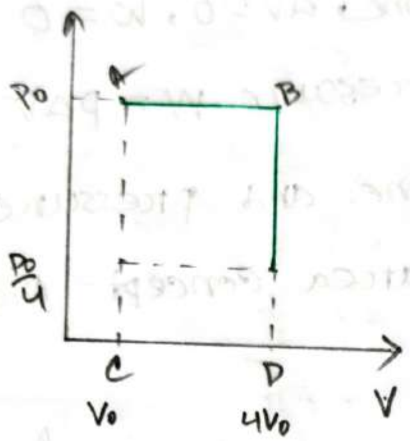
(a)

Since, ~~done~~ work done equals the area under the pv curve.

so, for A path:—



here, $V_0 = 1 m^3$

$4V_0 = 4 \times 1 = 4 m^3$

$P_0 = 40 \, Pa$

$\frac{P_0}{4} = \frac{40}{4} = 10 \, Pa$

we know:—

$W = \text{Area of } ABCD$

$= AC \times CD$

$= 40 \times 3$

$= 120 \, J$

Ans!—

$AC = 40$

$CD = 4V_0 - V_0$

$= 4 - 1$

$= 3$

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Alternative:-

(a)

We know:-

The total work done by any gas is:-

$$W = \int_{V_i}^{V_f} P \cdot dV$$

$$= P(V_f - V_i)$$

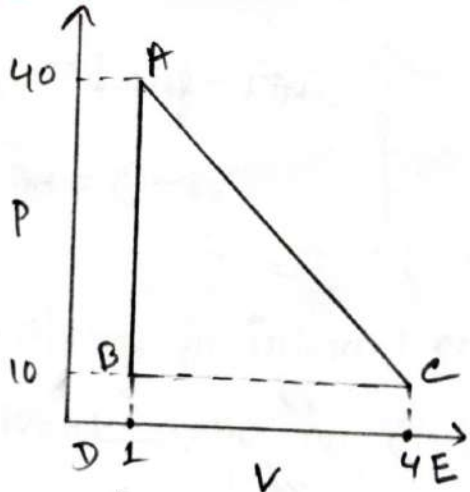
$$= 40 \times (4 - 3)$$

$$= 120 J$$

$P = 40$ pa
$V_f = 4\ m^3$
$V_i = 1\ m^3$

(b)

For B Path:-



We know:-

$$W = \text{Area of } ABC + \text{Area of } BCDE$$

$$= \frac{1}{2} \times AB \times BC + BC \times BD$$

$$= \frac{1}{2} \times 30 \times 3 + 3 \times 10$$

$$= 75 J$$

$AB = 40 - 10 = 30$
$BC = 4 - 1 = 3$
$BD = 10 - 0 = 10$

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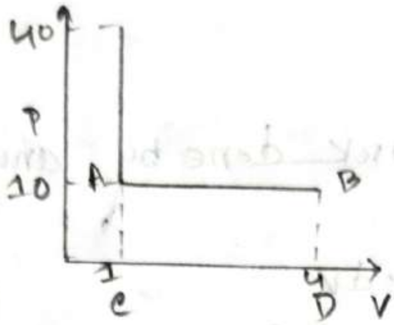
(c)

For c Path:—



we know:—

Work $W = \text{Area of } ABCD$

$W = AB \times AC$

$= 3 \times 10$

$= 30\,J$

Ans:—

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The First Law of thermodynamics:-

⇒ The first law of thermodynamics states that heat is a type of energy, and in thermodynamic processes, energy is always conserved. This means that heat energy can't be create or destroyed. It can move from one place to another and change into diffrent type of energy, and vice versa.

In simple terms, energy in the form of heat is never lost, but it can be transformed and move around.

⇒ The first Law of Thermodynamics Equation:-

$$\boxed{\Delta E_{int} = E_{int.f} - E_{int.i}}$$

$$\therefore \boxed{\Delta E_{int} = Q - W}$$

ΔE = Change in internal energy of the system
W = Work done on or by the system
Q = heat transferred to or from the system

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Points to remember:-

1. For an isolated system, energy (E) always remains constant.

2. For an ideal gas, the internal energy is a function of temperature only.

Some special Cases of the First Law of thermodynamics:-

1. Adiabatic processes:- (রুদ্ধতাপ প্রক্রিয়া)

⇒ Adiabatic processes involve no heat transfer and result in changes in temperature, pressure, and properties due to work. They are fast and have Applications in energy transfer and insulation.

That means; $Q = 0$

So,

$$\Delta E_{int} = Q - W$$

$$= 0 - W$$

$$\boxed{\Delta E_{int} = -W} \text{ [adiabatic process]}$$

→ if work is done by the system:- (that is, if W is positive)

$$\Delta E_{int} = -(+W)$$

$$\boxed{\Delta E_{int} = -W}$$

so, the internal energy of th system is Decreases

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→ if work is done on the system (that means ω is negative) the internal energy of the system Increases by that amount.

$$\Delta E_{int} = -(-W)$$

$$\boxed{\Delta E_{int} = +W}$$

(2) Constant-Volume processes:- (সমআয়তন প্রক্রিয়া):-

⇒ If the volume of a system is held constant, it's called Constant-Volume processes.

that means: $dv = 0$

We know:-

$$W = P \cdot dv$$
$$= P \times 0$$
$$= 0$$

$$\therefore \Delta E_{int} = Q - W$$
$$= Q - 0$$

$$\boxed{\Delta E_{int} = Q}$$

So, we can say that the system can do no work.

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→ If heat is absorbed by a system (that is, if Q is positive) The internal energy of the system Increases.

$$\boxed{\Delta E_{int} = +Q}$$

→ If heat is lost during the process (that is, if Q is negative) the internal energy of the system Decreases.

$$\boxed{\Delta E_{int} = -Q}$$

③ Cyclical processes:—

⇒ The processes in which the inital and final states are the same is known as cyclical processes. It's a Sequnce of processes, that leave the system in the same state in which is started.

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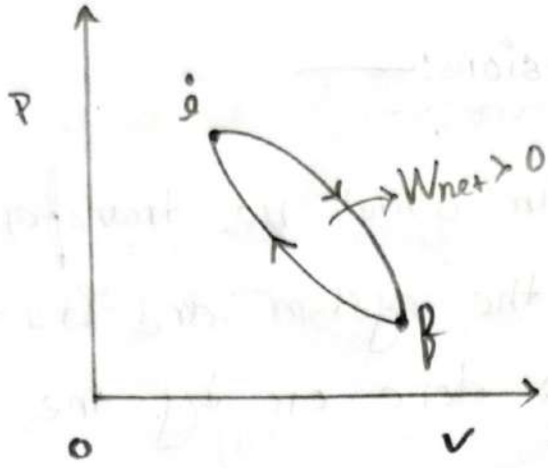

here, $E_i = E_f$

so, $\Delta E_{int} = E_i - E_f = E_f - E_i = 0$

we know—

$$\Delta E_{int} = Q - W$$
$$0 = Q - W$$
$$\therefore \boxed{Q = W}$$

→ The net work done during the process must exactly equal the net amount of energy transferred as heat. The store of initial energy of the system remains unchanged.

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(4.) Free Expansions:—

⇒ The process in which no transfer of heat Occurs between the system and its environment. And no work is done on by the system.

that means:—

$$Q = W = 0$$

we know:—

$$\Delta E_{int} = Q - W$$

$$\boxed{\Delta E_{int} = 0}$$

Problem 46:—

⇒ Suppose 200J of work is done on a system and 70 cal is extracted from the system as heat. In the sense of the first law of thermodynamics, what are the values of

(a) W

(b) Q

(c) ΔE_{int}

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(a)

⇒ Since the work is done on a system. That means the work done is negative.

$\therefore W = -200\,J$

(b)

⇒ Since heat is extracted from the system. That means Q is negative.

$\therefore Q = -70$ cal

$= -70 \times 4.2$ [1 cal = 4.2 J]

$= -294\,J$

(c)

We know:—

$\Delta E_{int} = Q - W$

$= -294 - (-200)$

$= -94\,J$

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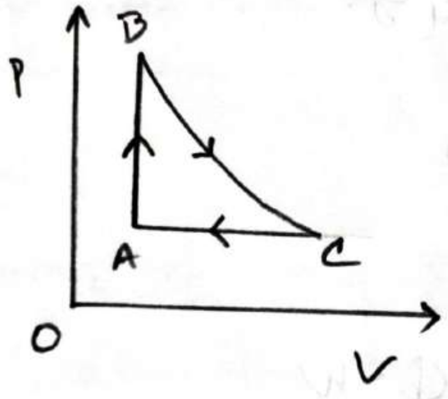
Problem : 48 :—

⇒ As a gas is held with in a closed chamber, it Passes through the cycle shown in the Fig. Determine the energy transferred by the system as heat during constant pressure system CA. if the energy added as heat Q_{AB} during constant-Volume processes, AB = 20 J. No energy is transferred as heat during adabatic processes BC. and the net work done during the cycle is 15 J



⇒ is given :—

$W = 15J$

$Q_{AB} = 20J$

$Q_{BC} = 0J$

For a cycle processes $\Delta E_{int} = 0$

We know:—

$$\Delta E_{int} = Q - W$$

$$Q = W$$

$$\therefore \; Q_{AB} + Q_{BC} + Q_{CA} = W$$

$$Q_{CA} = W - Q_{AB} - Q_{BC}$$

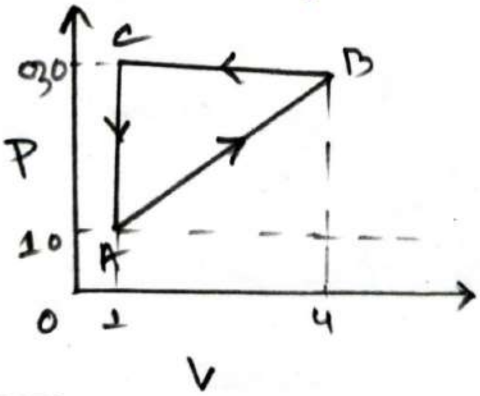
$$= 15 - 20 - 0$$

$$= -5 \text{ J}$$

Ans:-

Q.HW:—

⇒ A gas with a closed chamber undergoes the cycle shown in the Fig. The Horizontal scale is set by $V_s = 4m^3$. Calculate the net energy added to the system as heat during one complete system.



⇒ is given:—

$$V_s = 4m^3$$

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For a cyclical processes $\Delta E_{int} = 0$

We know:-

$$\Delta E_{int} = Q - W$$

$$0 = Q - W$$

$$Q = W$$

$$= -\frac{1}{2}(AC \times BC)$$

$$= -\frac{1}{2}(30-10) \times (4-1)$$

$$= -30 J$$ [Since Anticlockwize]

⇒ This means that 30 J of heat energy has been released by the system. or 30 J of heat is lost during the process.

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The Kinetic Theory of Gases

Ch: 19

#Avogadro's number, NA:-

(1.) 1 mole of a Substance Contains,

$$\boxed{N_A = 6.02 \times 10^{23}}$$

(2.) 1 mole of He Contains, $N_A = 6.02 \times 10^{23}$ atoms

∴ n mole He Contains.

Total number of atoms $N = nN_A$ atoms

(3.) 1 mole of O_2 Contains, $N_A = 6.02 \times 10^{23}$ molecules

∴ n mole of O_2 Contains,

Total number of molecules, $N = nN_A$ molecules

(4) 1 molar mass, $M = mN_A$

M = molar mass

m = mass of 1 atom or molecule

$N_A = 6.023 \times 10^{23}$

∴ n molar mass $M_s = m(nN_A) = n(mN_A)$

$M_s = mN$

$M_s = Mn$ → 1 molar mass

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(5) number of moles,

$$\boxed{n = \frac{m_s}{M} = \frac{n(mN_A)}{mN_A}} \boxed{= \frac{N}{N_A}}$$

$$n = \frac{n \text{ molar mass/Sample mass}}{1 \text{ molar mass}}$$

Here,

$N_A = 6.02 \times 10^{23}$

N = total number of atoms/molecule

n = number of mole per sample

m = mass per atom/molecule

m_s = Sample mass

M = Molar Mass

Q: The mass of a sample of H_2 gas is 5 g and the mass of its one molecule is 3.32×10^{-27} kg. Calculate its,

(1) Molar mass

(2) Number of mole

(iii) Total number of molecules.

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⇒ is given:—

$m_s = 50 \times 10^{-3} kg$

$m = 3.32 \times 10^{-27} kg$

1

We know:—

Molar Mass $M = mN_A$

$= 3.32 \times 10^{-27} \times 6.02 \times 10^{23}$

$= 1.99 \times 10^{-3}$

2

We know:—

Number of mole $n = \frac{m_s}{M}$

$= \frac{50 \times 10^{-3}}{1.99 \times 10^{-3}}$

$= 25.12$

3

We know:—

Total Number of molecules $N = nN_A$

$= 25.12 \times 6.023 \times 10^{23}$

$= 1.513 \times 10^{25}$

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Ideal gases:—

⇒ An Ideal gas is a gas in which the particles do not attract or repel one another and take up no space.

ideal gas Law:—

$$\boxed{PV = nRT}$$

P = pressure.
n = number of moles.
T = Temperature in kalvin
$R = 8.31\ J mol^{-1} K^{-1}$

Alternative form of ideal gas Law:—

We know:—

Boltzman Constant $\boxed{k = \frac{R}{N_A} = 1.38 \times 10^{-23}}$

and $n = \frac{N}{N_A}$

$R = kN_A$

$\therefore PV = nRT$
$= \frac{N}{N_A} \times kN_A \cdot T$

$$\boxed{PV = NkT}$$

N = Number of atoms/Molecule
k = Boltzman Constant

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#Work done by an Ideal gas at Constant temperature:-

We know:-

$$W = \int_{v_i}^{v_f} p\,dv \quad \text{———} \quad (1)$$

$$PV = nRT$$

$$\therefore p = \frac{nRT}{V} \quad \text{———} \quad (ii)$$

Substitude the value of p in the equation (1)

$$\therefore W = \int_{v_i}^{v_f} \frac{nRT}{V} \cdot dv$$

$$= nRT \int_{v_i}^{v_f} \frac{1}{v} \cdot dv$$

$$= nRT \cdot [\ln v]_{v_i}^{v_f}$$

$$= nRT * (\ln v_f - \ln v_i)$$

$$\boxed{W = nRT * \ln \frac{v_f}{v_i}}$$

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Some Condition:-

① For an expansion, $\boxed{V_f > V_i}$, W is $\boxed{\text{positive}}$.

② For an compression, $\boxed{V_i > V_f}$, W is $\boxed{\text{negative}}$

③ At Constant Volume:

$$V_f = V_i,\ \ln\frac{V_f}{V_f} = \ln\frac{1}{1} = \ln 1 = 0$$

$$\boxed{\therefore W = 0}$$

④ At Constant pressure:-

$$W = P\int_{V_i}^{V_f} dV$$

$$\boxed{W = P\Delta V} \qquad \Delta V = V_f - V_i$$

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Problem 4:—

⇒ A quantity of ideal gas at $10°C$ and $100 kPa$ Occupies a volume of $2.50 m^3$

(a) How many moles of the gas are present.

(b) If the pressure is raised to $300 kPa$ and the temperature is raised to $30°C$, how much volume does the gas occupy? Assume no leaks.

(a)

⇒ In given:—

$$T = 10°C = 283K$$
$$P = 100 kPa = 100 \times 10^3 Pa$$
$$V = 2.50 m^3$$
$$R = 8.31 \ J/mol\text{-}K$$

we know:—

$$PV = nRT$$
$$\therefore n = \frac{PV}{RT}$$
$$= \frac{100 \times 10^3 \times 2.50}{8.31 \times 283}$$
$$= 106.30 \ mol$$

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(b)

In given:-

$T_1 = 10^\circ C = 283\,K$

$P_1 = 100\,kPa = 100 \times 10^3\,Pa$

$V_1 = 2.50\,m^3$

$T_2 = 30^\circ C = 303\,K$

$P_2 = 300\,kPa = 300 \times 10^3\,Pa$

$V_2 = ?$

We know:-

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\therefore V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$= \frac{\cancel{283} \times 2.50 \times 303 \times 100 \times 10^3}{300 \times 10^3 \times 283}$$

$$\therefore V_2 = 0.89\,m^3$$

Ans:-

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Problem: 7

⇒ Suppose 1.80 mol of an ideal gas is taken from a volume of $3 m^3$ to a volume of $1.50 m^3$ via an isothermal compression at 30°c

(a) How mouch energy is transferred as heat during the compression.

(b) Is the transfer to or from the gas?

⇒

(a)

Since it is a Isothermal process, from the 1st Law of the thermodyamies, change in internal Energy (ΔE_{int}) is zero.

We know:-

$\Delta E_{int} = Q - W$

$\Rightarrow 0 = Q - W$

$\Rightarrow Q = W$ ——— (1)

Again:-

is given:-

$n = 1.80 mol$

$V_i = 3m^3$

$V_f = 1.50 m^3$

$T = 30°C = 303K$

We know:-

$W = nRT \ln \frac{V_f}{V_i}$

$= 1.80 \times 8.31 \times 303 \times \ln \frac{1.5}{3} = -3141.53 J$

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Substitude the value of W in the equation (1)

$\therefore Q = -3141.53\ J$

(b)

Since, we have got negative amount of heat, that means heat is transferred from the gas.

Problems for practice:-

(1) Find the mass in kilograms of 7.50×10^{24} atoms of arsenic, which has a molar mass of 74.9 g/mol.

$\Rightarrow$ In given:-

$N = 7.50 \times 10^{24}$ atoms

$M_g = 74.9 \times 10^{-3}$ kg/mol

$m = ?$

We know:-

$M = m N_A$

$\therefore m = \dfrac{M}{N_A}$

$= \dfrac{74.9 \times 10^{-3}}{6.023 \times 10^{23}}$

$= 1.24 \times 10^{-25}$ kg

$\therefore$ Total mass $= m \times N = 0.93$ kg

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(2) Gold has a molar mass of 197 g/mol.

(a) How many moles of gold are in a 2.50 g sample of pure Gold.

(b) How many atoms are in the sample.

(a)

⇒ As given:—

$$m = 197\ g/mol = 197 \times 10^{-3}\ Kg/mol$$

$$m_s = 2.50g = 2.50 \times 10^{-3}\ Kg.$$

We know:—

$$n = \frac{m_s}{m}$$

$$= \frac{2.50 \times 10^{-3}}{197 \times 10^{-3}}$$

$$= 0.0127\ mole$$

(b)

As given:—

$$n = 0.0127\ mole$$

$$N_A = 6.023 \times 10^{23}$$

We know:—

$$N = nN_A$$

$$= 0.0127 \times 6.023 \times 10^{23}$$

$$= 7.64 \times 10^{21}$$

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(1)

⇒ Find the mass in kilograms of 7.50×10^{24} atoms of arsenic, which has a molar mass 74.9 g/mol

⇒ In given:-

$N = 7.50\times10^{24}$ atoms

$N_A = 6.023\times10^{23}$

$M = 74.9$ g/mol $= 74.9\times10^{-3}$ kg/mol

We know-

$$n = \frac{N}{N_A}$$

$$= \frac{7.50\times10^{24}}{6.023\times10^{23}}$$

$$= 12.453 \text{ moles.}$$

Again, we know:-

$$M = mN_A$$

$$\therefore m = \frac{M}{N_A} = \frac{74.9\times10^{-3}}{6.02\times10^{23}}$$

$$= 1.24\times10^{-25} \text{ kg}$$

m = mass of 1 atom

$\therefore$ Total mass $= m\times N$

$= 1.24\times10^{-25}\times7.50\times10^{24}$

$= 0.93$ kg

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(3)

⇒ O_2 gas having a volume of $1000 cm^3$ at $40°C$ and $1.01 \times 10^5 Pa$ expands untill its volume is $1500 cm^3$ and its pressure is $1.06 \times 10^5 Pa$. Find

(a) The number of moles of O_2 present and

(b) The final temperature of the sample.

(a)

is given:—

$$P = 1.01 \times 10^5 Pa$$

$$V = 1000 cm^3 = 1000 \times 10^{-6} m^3$$

$$T = 40°C = 313 K$$

we know —

$$PV = nRT$$

$$\therefore n = \frac{PV}{RT}$$

$$= \frac{1.01 \times 10^5 \times 1000 \times 10^{-6}}{8.31 \times 313}$$

$$= 0.039 \text{ moles}$$

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(b)

is given:—

$P_1 = 1.01 \times 10^5 Pa$

$T_1 = 40°c = 313k$

$V_1 = 1000 cm^3 = 1000 \times 10^{-6} m^3$

$V_2 = 1500 cm^3 = 1500 \times 10^{-6} m^3$

$T_2 = ?$

$P_2 = 1.06 \times 10^5 Pa$

we know:—

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Rightarrow T_2 = \frac{T_1P_2V_2}{P_1V_1}$$

$$= \frac{1.06 \times 10^5 \times 313 \times 1500 \times 10^{-6}}{1.01 \times 10^5 \times 1000 \times 10^{-6}}$$

$$= 492.74 K$$

∴ The final Temparature is 492.74 K

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(B)

⇒ The best laboratory vacuum has a pressure of about 1.00×10^{-18} atm, or 1.01×10^{-13} Pa. How many gas molecules are there per cubic centimeter in such a vacuum at 223K

⇒ In given:—

$P = 1.01 \times 10^{-13}$ Pa

$V = 1 \times 10^{-6} m^3$

$T = 223K$

$R = 8.31 J mol^{-1} K^{-1}$

We know:—

$PV = nRT$

$$\therefore n = \frac{PV}{RT}$$

$$= \frac{1.01 \times 10^{-13} \times 1 \times 10^{-6}}{8.31 \times 223}$$

$$= 41.48 \times 10^{-24} \text{ mole.}$$

We know.—

The total number of molecules $N = nN_A$

$= 41.48 \times 10^{-24} \times 6.023 \times 10^{23}$

$= 24.98$ molecules.

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(8.)

⇒ Compute:

(a) the number of moles and

(b) the number of molecules in 1.00 cm^3 of an ideal gas at a pressure of about 100 Pa and temperature of 220K.

(a)

⇒ is given:—

$V = 1 cm^3 = 1 \times 10^{-6} m^3$

$P = 100 Pa$

$T = 220K$

$R = 8.31 J mol^{-1} K^{-1}$

we know:—

$PV = nRT$

$n = \frac{PV}{RT}$

$= \frac{100 \times 1 \times 10^{-6}}{8.31 \times 220}$

$= 5.45 \times 10^{-8} mol.$

(b)

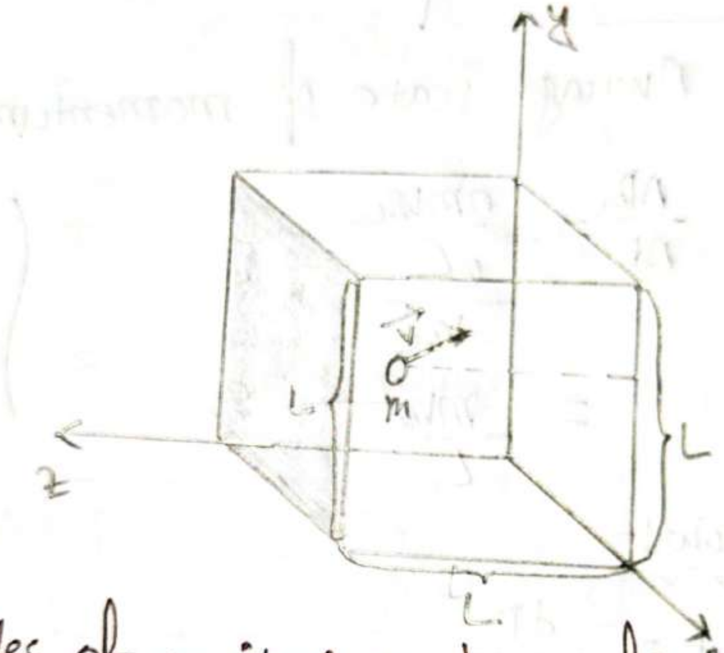
we know:—

The total number of Molecules $N = n \times N_A$

$= 5.45 \times 10^{-8} \times 6.023 \times 10^{23}$

$= 3.28 \times 10^{16}$

Relationship between:— Pressure, temperature and Rms speed:—



$\Rightarrow$ Let, n moles of an ideal gas be confined in a cubical box of volume $V = L^3$ at temperature T.

N number of molecules are moving all direction (x, y, z) with diffrent speeds $(v_1, v_2, \dots v_n)$ in the box. Consider only elastic collision with the walls of area L^2.

The change in momentum (ভরবেগ) of a molecule due to a collision with a wall in One dimension (Along x-axis)

$$\Delta P_x = P_f - P_i = -mv_x - mv_x$$
$$= -2mv_x$$

$$\left\{ \begin{aligned} P_i &= mv_x \\ P_f &= -mv_x \end{aligned} \right.$$

So, the momentum delivered by one molecule to the wall during the collision is

$$\boxed{\Delta P_x = 2mV_x}$$

and the Average rate of momentum transfer

$$\frac{\Delta P_x}{\Delta t} = \frac{2mV_x}{\frac{2L}{V_x}}$$

$$= \frac{mV_x^2}{L}$$

$$\left(\begin{array}{l} S = Vt \\ t = \frac{S}{V} \\ = \frac{2L}{V_x} \end{array} \right.$$

We know:-

$$F = \frac{dP}{dt}$$

$$\therefore F_x = \frac{mV_{x1}^2}{L} + \frac{mV_{x2}^2}{L} + \frac{mV_{x3}^2}{L} + \cdots + \frac{mV_{xn}^2}{L}$$

$$= \frac{m}{L}\left(V_{x1}^2 + V_{x2}^2 + V_{x3}^2 + \cdots + V_{xn}^2\right)$$

$$= \frac{mN}{L}\left(\frac{V_{x1}^2 + V_{x2}^2 + V_{x3}^2 + \cdots V_{xn}^2}{N}\right)$$

$$= \left(\frac{m n N_A}{L}\right) * (V_x^2)_{Avg} \cdots\cdots (1)$$

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∴ The pressure generated by all the molecules on the wall is

$$P = \frac{F_x}{L^2}$$

$$= \frac{\left(\frac{m n N_A}{L}\right) \times (V_x)^2_{AVG}}{L^2}$$

$$= \frac{n m N_A \times (V_x)^2_{AVG}}{L^3} \quad [M = mN_A]$$

$$= \frac{nM \times (V_x)^2_{AVG}}{V}$$

$$\therefore \boxed{P = \frac{nM \times (V_x)^2_{AVG}}{V}}$$

M = molar mass

V = Volume of the gas

Now :-

$$\therefore (V_x)^2_{AVG} = \frac{PV}{nM}$$

$$= \frac{nRT}{nM}$$

$$\boxed{(V_x)^2_{AVG} = \frac{RT}{M}} \quad \ldots\ldots\ldots (1)$$

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For N molecules (any gas, ideal or real):- (x, y, z-axes)

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

$$\Rightarrow (v^2)_{avg} = (v_x)^2_{Avg} + (v_y)^2_{Avg} + (v_z)^2_{Avg}$$

$$\Rightarrow v^2_{Avg} = (v_x)^2_{Avg} + (v_x^2)_{Avg} + (v_x)^2_{Avg} \quad [v_x = v_y = v_z]$$

$$\Rightarrow (v^2)_{Avg} = 3(v_x)^2_{Avg}$$

$$\therefore \boxed{(v_x)^2_{Avg} = \frac{1}{3}(v^2)_{Avg}}$$

Substitude the value of $(v_x^2)_{Avg}$ in the equation (I)

$$\frac{RT}{M} = \frac{1}{3}(v^2)_{Avg}$$

$$\boxed{\therefore v^2_{Avg} = \frac{3RT}{M}}$$

Root Mean Square Velocity:-

⇒ the sequare root of $(v^2)_{Avg}$ is kind of average speed, called the root-mean-square speed of the molecules and symbolized by V_{rms}.

⇒ The V_{rms} Velocity is directly proportional to the square root of temperature and inversely pro-portional to the square root of molar mass.

$$\therefore V_{rms} = \sqrt{V^2_{avg}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{V_1^2+V_2^2+V_3^2+\cdots+V_n^2}{n}}$$

Check point:-

⇒ Five gas molecules chosen at random are found to have speeds of 500, 600, 700, 800 and 900 m/s.

(a) what is the average speed

(b) what is the rms speed.

⇒

(a)

$$V_{avg} = \frac{500+600+700+800+900}{5}$$

$$= 700\ ms^{-1}$$

Ans!

(b)

in given:—

$$V_{Avg} = 700\ ms^{-1}$$

we know:—

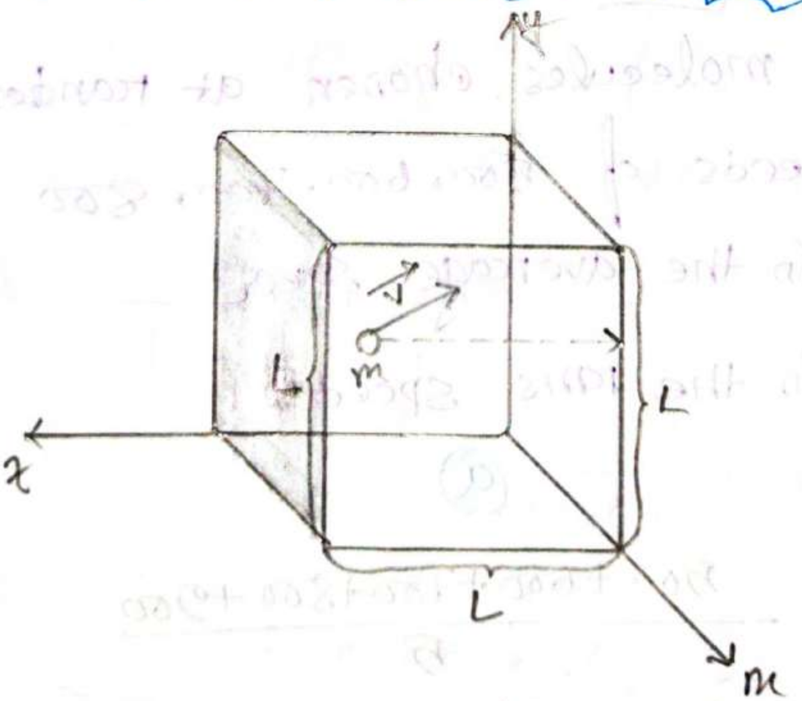
$$V_{rms} = \sqrt{v_1^2+v_2^2+v_3^2+v_4^2+v_5^2/5}$$

$$V_{rms} = \sqrt{\frac{v_1^2+v_2^2+v_3^2+v_4^2+v_5^2}{5}}$$

$$= \sqrt{\frac{(500)^2+(600)^2+(700)^2+(800)^2+(900)^2}{5}}$$

$$= 714.14\ ms^{-1}$$

Translational Kinetic Energy:—



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⇒ Now, we consider a single molecule of an ideal gas. Its moves around in the box but we now assume that its speed changes when it collides with other molecules. It is translational kinetic energy, at any instant $\frac{1}{2}mv^2$.

So, Average translational kinetic energy over the time,

$$K_{avg} = \left(\frac{1}{2}mv^2\right)_{avg}$$

$$= \frac{1}{2}m V^2_{avg}$$

$$= \frac{1}{2}m V^2_{rms} \qquad \left[V_{rms} = \sqrt{V^2_{avg}} = \sqrt{\frac{3RT}{M}} \right]$$

$$= \frac{1}{2}m \times \frac{3RT}{M} \qquad \left[M = mN_A \right]$$

$$= \frac{3}{2} \times \frac{m}{M} \times RT \qquad \left[\frac{1}{N_A} = \frac{m}{M} \right]$$

$$= \frac{3}{2} \times \frac{1}{N_A} \times RT$$

$$\boxed{K_{avg} = \frac{3}{2}KT} \qquad \left[K = \frac{R}{N_A} \right]$$

K = Boltzmann Constant $= 1.38 \times 10^{-23}$ J/K

T = Temperature (K)

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Problem 18:—

⇒ The temperature and pressure in the Sun's atomosphere are 2.00×10^{6} K and 0.0300 Pa. Calculate the rms speed of free electrons (mass 9.11×10^{-31} kg) there, assuming they are an ideal gas.

⇒ in given:—

$$T = 2\times10^{6}\ K$$
$$P = 0.0300\ Pa$$
$$m = 9.11\times10^{-31}\ kg$$

We know:—

$$M = mN_A$$
$$= 9.11\times10^{-31}\times6.023\times10^{23}$$
$$= 5.486\times10^{-7}\ kg$$

Now:— we know:—

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$
$$= \sqrt{\frac{3\times8.31\times2\times10^{6}}{5.486\times10^{-7}}}$$
$$= 9.533\times10^{6}\ ms^{-1}$$

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problem: 25:-

⇒ Determine the average value of the translational kinetic energy of the molecule of an ideal gas at temperature.

ⓐ 0.00°c

ⓑ 100°c. what is the translational kinetic energy per mole of an ideal gas at,

ⓒ 0.00°c

ⓓ 100°c

⇒

ⓐ

is given:-

$T = 0°c = 273K$

We know:-

$$K_{avg} = \frac{3}{2} KT$$

$$= \frac{3}{2} \times 1.38 \times 10^{-23} \times 273$$

$$= 5651.1 \times 10^{-24} J$$

ⓑ

in given:- $T = 100°c = 373K$

We know:- $K_{avg} = \frac{3}{2} KT$

$$= \frac{3}{2} \times 1.38 \times 10^{-23} \times 373$$

$$= 7721.1 \times 10^{-24} J$$

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(c)

is given:—

$$Kavg_2 = 7721.10 \times 10^{-24} J$$

$$kavg_1 = 5651.1 \times 10^{-24} J$$

We know:—

Kavg Per mole = $Kavg_1 \times N_A$

$= 5651.1 \times 10^{-24} \times 6.023 \times 10^{23}$

$= 3404 J$

(d)

We know:—

Kavg Per mole = $Kavg_2 \times N_A$

$= 6.023 \times 10^{23} \times 7721.10 \times 10^{-24}$

$= 4650.41 J$

Ans:—

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#Degrees of Freedom & Equipartition of Energy :-

Degrees of Freedom :- (স্বাধীনতার মাত্রা)

⇒ Degress of Freedom are the number of independent Variable that can be estimated in a statistical analysis.

Type	Example	Translational DoF	Rotational DOF Bond	Total DOF D.f
Monatomic	He N=1	3	0	$D.f = 3N - B$ $= 3\times1 - 0$ $= 3$
Diatomic	N=2	3	1	$Df = 3N - B$ $= 3\times2 - 1$ $= 5$
Polyatomic	N=3	3	3	$D.f = 3N - B$ $= 3\times3 - 3$ $= 6$

Equipartition of Energy :-

⇒ States that molecules in thermal equilibrium have the same average energy associated with eaten independent degree of freedom of their motion and that the energy is per molecule or per mole.

$$\boxed{KE = \frac{1}{2}KT}$$

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Internal Energy of an Ideal gas:

Let, Consider a monatomic gas.

we define the enternal energy as the sum of the translational kinetic energy of atoms.

$$\therefore E_{int} = N k_{avg}$$

$$= n N_A \times \left(\frac{3}{2} kT\right)$$

$$= \frac{3}{2} \times N_A k \times nT$$

$$\boxed{E_{int} = \frac{3}{2} nRT}$$

$k = \frac{R}{N_A}$

$\therefore R = N_A \times k$

For change in temperature : ΔT

$$\boxed{\Delta E_{int} = \frac{3}{2} nR\Delta T}$$

In general the change in internal energy of an Ideal gas:—

$$\boxed{\Delta E_{int} = \frac{f}{2} nR\Delta T}$$

f = total degrees of freedom.

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Molar Specific Heat of an Ideal Gas:—

We know:—

$$C_m = \frac{Q}{n\Delta T}$$

$$\therefore Q = C_m n \Delta T$$

There are two types of molar specific heat:—

1. Molar specific heat at constant volume (C_v)
2. Molar specific heat at constant pressure (C_p)

Molar Specific Heat at Constant Volume C_v

Added amount of heat $Q = nC_v\Delta T$

Work done $W = 0$

$$\left\{ \begin{aligned} [\,W &= P\Delta V\,] \\ &= P(V_f - V_i) \\ &= P[(V_f - V_f)] \quad [V_f = V_i] \\ &= P \times 0 \\ &= 0 \end{aligned} \right\}$$

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we know—

$$\Delta E_{int} = Q - W$$

$$\boxed{\Delta E_{int} = nC_v\Delta T - 0}$$

$$\therefore C_v = \frac{\Delta E_{int}}{n\Delta T} \qquad \left[\Delta E_{int} = \frac{Df}{2} \times nR\Delta T\right]$$

$$\therefore C_v = \frac{\frac{Df}{2} nR\Delta T}{n\Delta T}$$

$$\boxed{C_v = \frac{Df}{2} \times R}$$

∴ Value of C_v for monatomic ideal Gas:—

$$\boxed{C_v = \frac{3}{2}R} \qquad D.f = 3$$

∴ Value of C_v for Diatomic ideal Gas:—

$$\boxed{C_v = \frac{5}{2}R} \qquad D.f = 5$$

∴ Value of C_v for polyatomic ideal Gas:—

$$\boxed{C_v = \frac{6}{2}R} \qquad D.f = 6$$

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Molar Specific Heat at Constant Pressure, C_P

Added amount of heat, $Q = nC_p \Delta T$

work done, $W = p\Delta V$

$$\boxed{W = nR\Delta T} \quad [PV = nRT]$$

We know:—

$$\Delta E_{int} = Q - W$$

$$nC_V\Delta T = nC_p\Delta T - nR\Delta T \quad [\Delta E_{int} = nC_V\Delta T]$$

$$C_V = C_p - R$$

$$\therefore C_p = \frac{C_V + R}{1}$$

$$\therefore \boxed{C_p = C_V + R}$$

$C_p > C_V$ → All time.

∴ Value of C_p for monatomic ideal gas:—

$$\boxed{C_p = \frac{5}{2}R}$$

∴ Value of C_p for Diatomic ideal gas:—

$$\boxed{C_p = \frac{7}{2}R}$$

∴ Value of C_p for polyatomic ideal gas:—

$$\boxed{C_p = 4R}$$

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Problem: 47:—

⇒ The temperature of 2.00 mol of an ideal monatomic gas is raised 15K at Constant volume. what are.

(a) the work done by the gas,

(b) The energy transferred as heat Q,

(c) The change ΔE_{int} in the internal energy of the gas.

(d) The change ΔK_{avg} in the average kinetic energy per atom.

(a)

Is given:—

$n = 2$ mol

$D.f = 3$ for monatomic

$T = 15K$

$V_i = V_f$

we know—

$W = P\Delta V$

$= P \times (V_f - V_i)$

$= P \times (V_f - V_f)$

$= P \times 0$

$= 0J$

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(b)

we know:—

for Constant Volume $C_v = \frac{D.f}{2}.R$

$Q = nC_v\Delta T$

$= 2 \times \frac{3}{2} \times R \times \Delta T$

$= 3 \times 8.31 \times 15$

$= 373.95\,J$

(c)

we know:—

$\Delta E_{int} = Q - W$

$= 373.95 - 0$

$= 373.95\,J$

(d)

we know:—

$\Delta K_{avg} = \frac{3}{2} K\Delta T$

$= \frac{3}{2} \times (1.38 \times 10^{-23}) \times 15$

$= 3.105 \times 10^{-22}\,J$

$K = 1.38 \times 10^{-23}$

$\Delta T = 15K$

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Problem: 48:—

⇒ When 20.0J was added as heat to a particular ideal gas, the volume of the gas changed from 50 cm³ to 100 cm³ while the pressure remained 1 atm

(a) By how much did the internal energy of the gas change? If the quantity of gas present was 2.0×10^{-3} mol.

(b) Find cp.

(c) Find Cv.

⇒ is given:—

$$Q = 20.0 J$$
$$V_f = 100 cm^3 = 100\times10^{-6} m^3$$
$$V_i = 50 cm^3 = 50\times10^{-6} m^3$$
$$P = 1 atm = 101325 Pa$$

(a)

is given:—

$$n = 2.0\times10^{-3} mol$$

We know:—

$$\Delta E_{int} = Q - W$$
$$= 20.0 - P\Delta V$$
$$= 20.0 - P(V_f - V_i)$$
$$= 20.0 - 101325(100\times10^{-6} - 50\times10^{-6})$$
$$= 15.83 J$$

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(b)

Since, the pressure is constant

We know:—

$Q = nc_p\Delta T$

$c_p = \frac{Q}{n\Delta T}$ .... (i)

Again we know:—

$W = P\Delta V$

$P\Delta V = nR\Delta T$

$\therefore \Delta T = \frac{P\Delta V}{nR}$ .... (ii)

Substitute the value of ΔT in equation (i) $\Rightarrow$

$$c_p = \frac{Q}{n \times \frac{P\Delta V}{nR}}$$

$$= \frac{Q}{n} \times \frac{nR}{P\Delta V}$$

$$= \frac{QR}{P\Delta V}$$

$$= \frac{20.7 \times 8.31}{101325 \times 50 \times 10^{-6}}$$

$$= 34.28 \text{ J/mol-K}$$

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(c)

We know:—

$$C_p = C_v + R$$

$$\therefore C_v = C_p - R$$

$$= 34.28 - 8.31$$

$$= 25.97 \ J\,mol^{-1}\,K^{-1}$$

Problems for Practice:—

Problem: 42:—

What is the internal energy of 1 mol of an ideal monatomic gas at 273 K.

is given:—

$n = 1\,mol$

$\Delta T = 273\,K$

D.f = 3 for monatomic gas.

We know:—

The internal energy of an ideal gas

$$\Delta E_{int} = \frac{Df}{2} \times nR\Delta T$$

$$= \frac{3}{2} \times 1 \times 8.31 \times 273$$

$$= 3402.945 \ J$$

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Problem 43:—

⇒ The temperature of 3 mole of an diatomic gas is increased by 40.0°C without the pressure of the gas changing. The molecules in the gas rotate but do not oscillate.

ⓐ How much energy is transferred to the gas as heat.?

ⓑ What is the change in the internal energy of the gas?

ⓒ How much work is done by the gas?

ⓓ By how much does the rotational kinetic energy of the gas increase?

⇒ is given:—

$$\left.\begin{array}{l} n = 3 \text{ mol} \\ \Delta T = 40 K \end{array}\right\} D.f = 5 \text{ for Diatomic}$$

ⓐ

Since the pressure with out changing, that means the pressure is constant.

We know:—

$$Q = nc_p \Delta T$$
$$= 3 \times (c_v + R) \times \Delta T$$
$$= 3 \times \left(\frac{5}{2}R + R\right) \times 40 = 3490.2 J$$

$$\left\{\begin{array}{l} c_p = c_v + R \\ c_v = \frac{D.f}{2} \times R \end{array}\right.$$

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(b)

in given:—

D.f = 5 for Diatomic

$R = 8.31\ J mol^{-1} K^{-1}$

$\Delta T = 40K$, $n = 3$ mol

We know:—

The change in internal energy of an ideal gas:—

$$\Delta E_{int} = \frac{D.f}{2} nR\Delta T$$

$$= \frac{5}{2} \times 3 \times 8.31 \times 40$$

$$= 2493\ J$$

(c)

We know:—

$$W = P\Delta V$$

$$W = nR\Delta T$$

$$= 3 \times 8.31 \times 40$$

$$= 997.2\ J$$

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(d)

We have:—

$\Delta E_{int} = 2493\ J$

$W = 997.2\ J$

$Q = 3490.2\ J$

is given:—

D.f = 5 for Diatomic

$R = 8.31\ J mol^{-1} K^{-1}$

$\Delta T = 40\ K$

$n = 3\ mol$

We know:—

we know, Average translational kinetic energy:—

$K_{avg} = \frac{3}{2} kT$ → for single molecule.

Since, the rotational kinetic energy of the gas increase. so → for total number of molecule.

$\Delta K_{avg_t} = N(\Delta K_{avg})$

$= nN_A \times \frac{3}{2} \times \frac{R}{N_A} \times \Delta T$

$= n \times \frac{3}{2} R \times \Delta T$

$= \frac{3}{2} \times 8.31 \times 40 \times 3$

$= 1495.8\ J$

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Again we know.—

$$\Delta E_{int} = \Delta K_{avg_t} + \Delta K_{avg_{rot}}$$

$$\Delta K_{avg_{rot}} = \Delta E_{int} - \Delta K_{avg_t}$$
$$= 2423 - 1495.8$$
$$= 927.2 \text{ J}$$

Note:— If there been no rotation. All the energy would have gone into the translational kinetic energy.

Gamma (γ)

⇒ The specific heat ratio of a gas, is commonly defined as the ratio of the specific heat of the gas at a constant pressure to its specific heat at a constant volume.

So,
$$\boxed{\gamma = \frac{c_p}{c_v}}$$

Symbolized as gamma = γ

1. for monatomic gas $\gamma = 1.67$
2. for Diatomic gas $\gamma = 1.40$
3. for Polyatomic gas $\gamma = 1.33$

Relationship between Pressure and Volume in Adiabatic Process:—

We know:—

$$\Delta E_{int} = Q - W$$

$$\therefore \Delta E_{int} = 0 - W$$

$$nC_v \Delta T = -P\Delta V \quad \text{——} \quad (1)$$

$Q = 0$ for Adiabatic Process.

$\Delta E_{int} = nC_v \Delta T$

$W = P\Delta V$

Again, we know:—

$$PV = nRT$$

$$\Rightarrow \frac{d}{dT}(PV) = \frac{d}{dT}(nRT)$$

$$\Rightarrow P \times \frac{dV}{dT} + V \times \frac{dP}{dT} = nR$$

$$\Rightarrow \frac{1}{dT}(PdV + VdP) = nR$$

$$\Rightarrow PdV + V \cdot dP = nR \cdot dT$$

$$\Rightarrow dT = \frac{PdV + V \cdot dP}{nR} \quad \cdots\cdots \quad (ii)$$

Substitute the value of dT in equation (1) $\Rightarrow$

$$nC_v \times \frac{P \cdot dV + V \cdot dP}{nR} = -P \cdot dV$$

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$$\Rightarrow \frac{c_v(P \cdot dv + V \cdot dp)}{R} = -p \cdot dv$$

$$\Rightarrow c_v \cdot P \cdot dV + c_v \cdot VdP = -RP \cdot dV$$

$$\Rightarrow c_v PdV + c_v \cdot VdP + RP \cdot dV = 0$$

$$\Rightarrow c_v P \cdot dV + c_v \cdot VdP + (c_p - c_v) P \cdot dV = 0 \quad [C_p = c_v + R]$$

$$\Rightarrow \cancel{c_v PdV} + c_v \cdot VdP + C_p \cdot PdV - \cancel{c_v PdV} = 0$$

$$\Rightarrow c_v \cdot VdP + c_p \cdot PdV = 0$$

$$\Rightarrow VdP + \frac{c_p}{c_v} P \cdot dV = 0 \quad [\text{devided by } c_v]$$

$$\Rightarrow \frac{V \cdot dP}{PV} + \frac{\gamma P \cdot dV}{PV} = 0 \quad \left[\frac{c_p}{c_v} = \gamma\right] [\text{devided by } PV]$$

$$\Rightarrow \frac{dP}{P} + \frac{\gamma dV}{V} = 0$$

$$\Rightarrow \int \frac{dP}{P} + \int \frac{\gamma dV}{V} = 0$$

$$\Rightarrow \ln p + \gamma \ln V = C$$

$$\Rightarrow \ln p + \ln V^\gamma = C$$

$$\Rightarrow \ln(pV^\gamma) = C$$

$$\boxed{pV^\gamma = K}$$

$$\therefore \boxed{P_1 V_1^\gamma = P_2 V_2^\gamma}$$

Relationship between Volume and Temperature in Adibatic Process:—

We know:—

$$PV^{\gamma} = K \quad \text{——— (1)}$$

Again, we know:—

$$PV = nRT$$

$$\therefore P = \frac{nRT}{V}$$

Substitude the value of P in equation (1) ⇒

$$\frac{nRT}{V} \times V^{\gamma} = K$$

$$\Rightarrow TV^{\gamma-1} = \frac{K}{nR}$$ [Since, n, R also constant]

$$\Rightarrow \boxed{TV^{\gamma-1} = K_1}$$

$$\Rightarrow \boxed{T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}}$$

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Relationship between Volume and pressure in Adiabatic Process:—

We know:—

$$PV^{\gamma}=K \longrightarrow (1)$$

Again, we know:—

$$PV=nRT$$

$$V=\frac{nRT}{P}$$

Substitute the value of V in equation (1) $\Rightarrow$

$$P\left(\frac{nRT}{P}\right)^{\gamma}=K$$

$$\Rightarrow \frac{n^{\gamma}R^{\gamma}T^{\gamma}}{P^{\gamma}}\times P=K$$

$$\Rightarrow T^{\gamma}P^{1-\gamma}=\frac{K}{n^{\gamma}R^{\gamma}}$$

$$\Rightarrow \boxed{T^{\gamma}P^{1-\gamma}=K_1}$$

$$\Rightarrow \boxed{T_1P_1^{\frac{1-\gamma}{\gamma}}=T_2P_2^{\frac{1-\gamma}{\gamma}}}$$

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Work done for an ideal gas in an Adiabatic process:

We know,

$$PV^{\gamma} = K$$

$$\therefore P = \frac{K}{V^{\gamma}} \quad \text{--- (1)}$$

Again, we know

$$W = \int_{V_i}^{V_f} P\,dV$$

$$= \int_{V_i}^{V_f} \frac{K}{V^{\gamma}} \cdot dV$$

$$= K\left[\frac{V^{-\gamma+1}}{-\gamma+1}\right]_{V_i}^{V_f}$$

$$= \frac{K}{-\gamma+1}\left[V_f^{-\gamma+1} - V_i^{-\gamma+1}\right]$$

$$= \frac{P_f V_f^{\gamma} V_f^{-\gamma+1} - P_i V_i^{\gamma} V_i^{-\gamma+1}}{-\gamma+1}$$

$$= \frac{P_f V_f^{\gamma-\gamma+1} - P_i V_i^{\gamma-\gamma+1}}{-\gamma+1}$$

$$\boxed{W = \frac{P_f V_f - P_i V_i}{-\gamma+1}}$$

$$\boxed{W = \frac{nR}{-\gamma+1}\left(T_f - T_i\right)}$$

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Problem NO: 558 —

⇒ A certain gas occupies a volume of 4.3 L at a pressure of 1.2 atm and a temperature of 310 K. It is compressed adiabatically to a volume of 0.76 L. Determine.

(a) The final pressure.

(b) The final temperature, assuming the gas to be an ideal gas for which $\gamma = 1.4$

⇒ is given: —

$V_i = 4.3 L$

$P_i = 1.2 \text{ atm} = 1.2 \times 101325 \text{ Pa}$

$T_i = 310 K$

$\gamma = 1.40$

$V_f = 0.76$

(a)

We know: —

$P_i V_i^{\gamma} = P_f V_f^{\gamma}$

$$P_f = \frac{P_i V_i^{\gamma}}{V_f^{\gamma}}$$

$$= \frac{1.2 \times 101325 \times (4.3)^{1.40}}{(0.76)^{1.40}}$$

$$= 1.37 \times 10^6 \text{ Pa}$$

$$= 13.57 \text{ atm}$$

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(b)

we know:—

$$T_f V_f^{\gamma-1} = T_i V_i^{\gamma-1}$$

$$T_f = \frac{T_i V_i^{\gamma-1}}{V_f^{\gamma-1}}$$

$$= \frac{310 \times (4.3)^{1.4-1}}{(0.76)^{1.4-1}}$$

$$= 620.04 K$$

Problem: 54:—

⇒ We know that for an adiabatic process pV^γ = a Constant. Evaluate "a constant" for an adiabatic Process involving exactly 2.0 mol for an ideal gas Passing through the state having exactly $p = 1.0$ atm and $T = 300K$. Assume a diatomic gas whose molecules rotate but do not oscillate.

⇒ is given:—

$n = 2$ mol

$T = 300K$

$P = 1$ atm $= 101325$ Pa

we know:—

$$pV^\gamma = a \quad \text{—— (i)}$$

$$\gamma = \frac{C_P}{C_V} \quad \text{—— (ii)}$$

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$$C_V = \frac{D.f}{2} \times R$$

$$= \frac{5}{2} \times R = \frac{5}{2} \times 8.31 = 20.775$$

$$C_P = C_V + R$$

$$= \frac{5R}{2} + R$$

$$= \frac{7R}{2} = 29.085$$

Substitute the value of C_V & C_P in equation (ii) ⇒

$$\therefore \gamma = \frac{29.085}{20.775} = 1.40$$

We know:—

$$PV = nRT$$

$$V = \frac{nRT}{P}$$

Substitute the value of V in equation (i) ⇒

$$\therefore P\left(\frac{nRT}{P}\right)^{\gamma} = a$$

$$\therefore a = 101325 \times \left(\frac{2 \times 8.31 \times 300}{101325}\right)^{1.40}$$

$$= 1404.73 \ Nm^{2.2}$$

Ans.

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Ch: 10

Entropy and Second Law of Thermodynamics

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Irreversible process & Entropy:—

⇒ A process in which the system and the surroundings do not return to their orginal condition once the process in initiated.

* These one-way processes are irreversible, meaning that they cannot be reversed by means of only small changes in there enviroment.

* The key to understanding why one-way processes cannot be reversed involes a quantity known as entropy.

Entropy postulates—

⇒ If an irreversible process occurs in a closed system, the entropy S of the system always increase. It never decreases [$\Delta S > 0$]

* The energy of a closed system in conserved. It always remains constant, For irreversible processes, the entropy of a closed system always increases.

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Change in entropy:—

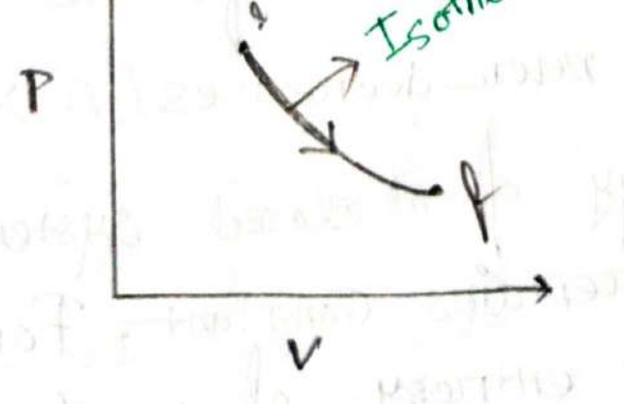
Let, the change in entropy $S_f - S_i$ of a system during a process that takes the system from an inital state i to final state f as

$$\boxed{\Delta S = S_f - S_i = \int_i^f \frac{dQ}{T}}$$

Q = the energy transferred as heat.
T = temperature of the system.

SI Unit = JK^{-1}

Reversible Isothermal Expansion:—



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⇒ To keep the temperature T of the gas constant during the isothermal Expansion. Heat Q must have been energy transferred from the reservoir to the gas. Thus Q is positive and the entropy increase during the isothermal process.

$$\therefore \boxed{\Delta S = \int_i^f \frac{dQ}{T} = \frac{1}{T}\int_i^f dQ = \frac{Q}{T}}$$

To find the entropy change for an irreversible process.

⇒ Replace that with any reversible process that connect the same initial and final state. then calculate the entropy change for this reversible process with

$$\boxed{\Delta S = \int_i^f \frac{dQ}{T}}$$

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Entropy as a State function:—

⇒ We can prove that entropy is a state function for the special and important case in which an ideal gas is taken through a reversible process.

We know:—

$$dE_{int} = dQ - dW$$

$$\therefore dQ = dE_{int} + dW \quad \text{———} \quad (I)$$

Again, we know:—

for reversible process $dw = pdv$

$dE_{int} = nc_v dT$

Substitute the value of dw and dE_{int} in equation (I) ⇒

$$\therefore dQ = nc_v dT + pdV \quad \text{———} \quad (II)$$

Again, we know:—

Ideal gas $pv = nRT$

$$\therefore p = \frac{nRT}{V}$$

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Substitute the value of p in equation (11) ⇒

$$\therefore dQ = nc_v dT + \frac{nRT}{V} dV$$

$$\frac{dQ}{T} = \frac{nc_v dT}{T} + nR\frac{dV}{V} \quad [\text{divided by } T]$$

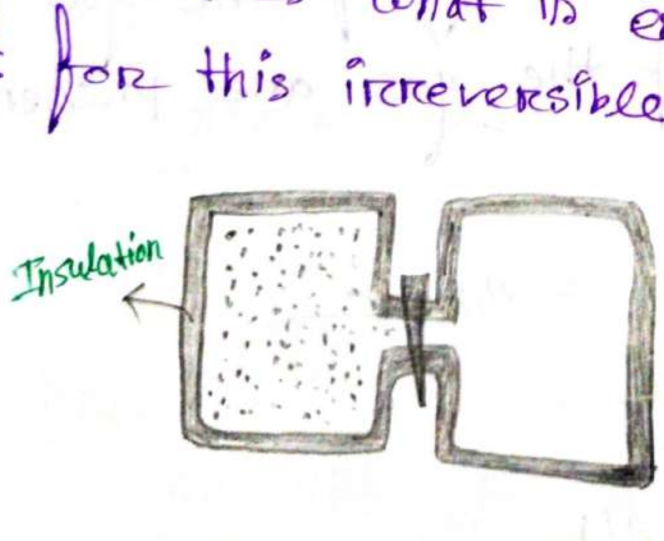
$$\int_i^f \frac{dQ}{T} = \int_i^f nc_v \frac{dT}{T} + \int_i^f nR\frac{dV}{V}$$

$$\Delta S = nc_v[\ln|T|]_i^f + nR[\ln|V|]_i^f$$

$$\therefore \boxed{\Delta S = nc_v \ln\frac{T_f}{T_i} + nR\ln\frac{V_f}{V_i}}$$

#Problem 20.02:—

⇒ Suppose 1 mole of nitrogen gas is confined to the left side of the container of Fig. 20.1a. You open the stopcock, and the volume of the gas doubles. What is entropy change of the gas for this irreversible process.



⇒ is given:—

$n = 1\,mol$

$V_i = V$

$V_f = 2V$

$T_i = T_f$

We know:—

$$\Delta S = nR\ln\frac{V_f}{V_i} + nc_v\ln\frac{T_f}{T_i}$$

$$= nR \times \ln\frac{2V}{V} + nc_v\ln\frac{1}{1}$$

$$= 1 \times 8.31 \times \ln\frac{2}{1} + 0$$

$$= 5.76 \quad J/K$$

Problem: 2:—

⇒ An ideal gas undergoes a reversible isothermal expansion at 77°C increasing its volume from 1.30L to 3.40L. The entropy change of the gas is 22.0 J/K How many moles of the gas are present.

⇒ Is given:—

$T_i = T_f = 77°C = 350K$

$V_i = 1.30L$

$V_f = 3.40L$

$\Delta S = 22\,J/K$

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we know:-

$$\Delta S = nR \ln \frac{V_f}{V_i} + n c_v \ln \frac{T_f}{T_i}$$

$$= nR \ln \frac{3.40}{1.30} + n c_v \ln \frac{350}{350}$$

$$22 = nR \times 0.96 + 0$$

$$\therefore n = \frac{22}{8.31 \times 0.96}$$

$$= 2.759 \text{ mol}$$

Ans:-

Problem: 3:-

⇒ A 2.5 mol Sample of an ideal gas expands reversibly and isothermally at 360K unit. Its volume is doubled. what is the increase in entropy of the gas?

⇒ is given:-

$n = 2.5$ mol

$T_f = T_i = 350K$

$V_i = V$

$V_f = 2V$

we know:-

$$\Delta S = nR \ln \frac{V_f}{V_i} + n c_v \ln \frac{T_f}{T_i}$$

$$= 2.5 \times 8.31 \times \ln \frac{2V}{V} + 0$$

$$= 14.40 J/K$$

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Problem: 4 :-

⇒ How much energy must be transferred as heat for a reversible isothermal expansion of an an ideal gas at 132°c if the entropy of the gas increases by 46 J/K.

⇒ is given:-

$T = 132°c = 405 K$

$\Delta S = 46 \ J/K$

We know:-

$\Delta S = \frac{Q}{T}$

$\therefore Q = \Delta S \times T$

$= 46 \times 405$

$= 18630 J$

Second Law of Thermodynamics in terms of Entropy : $\Delta S \geq 0$

⇒ The 2nd Law of thermodynamics state that any naturally occurring process will allways cause the universe's entropy (ΔS) to increase. The Law simply state that the entropy of an isolated system will never decrease over time.

Change in entropy in reversible processes:-

$$\Delta S = S_f - S_i = \int_i^f \frac{dQ}{T} = \frac{1}{T}\int_i^f dQ = \frac{Q}{T}$$

when T constant.

$$\boxed{\Delta S = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} = 0}$$

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

That means in this case the change in entropy is constant.

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⇒ If a process occurs in a closed system, the entropy of the system increases for irreversible processes ($\Delta S > 0$) and remains constant for reversible processes ($\Delta S = 0$). It never decreases.

Notes

⇒ In the real world almost all processes are irreversible processes.

Change In entropy in irreversible processes:—

$$\boxed{\Delta S = -\frac{Q}{T_1} + \frac{Q}{T_2}}$$

$T_1 > T_2 \rightarrow$ Always

$\Delta S > 0 \rightarrow$ Always.

$-\frac{dQ}{T_1}$ = Entropy loss of a heated object

$\frac{dQ}{T_2}$ = Increased entropy of a cold object

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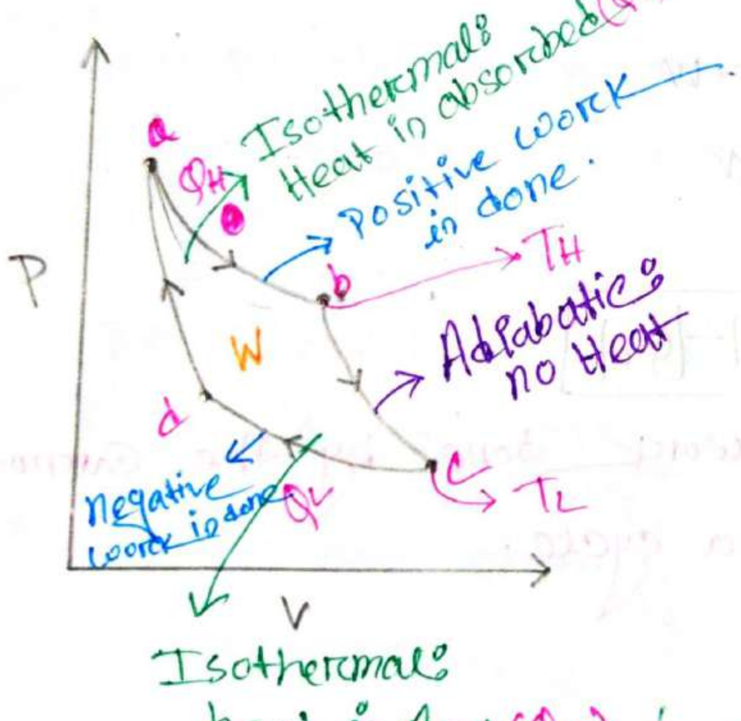
Heat Engine:-

⇒ An engine that converts thermal energy into mechanical energy is called a heat engine.

Carnot Engine:-

⇒ The Carnot engine is theoretical device that that determines the maximum efficiency of heat.



Here,
ab & cd = Isothermal Processes.

bc & da = adiabatic processes

T_H = High temperature

T_L = Low temperature.

Q_H = transferred heat from high temperature

Q_L = transferred heat from Low temperature.

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The Work :-

3) In a Carnot engine, the working substance completes reversible cycles. thus for a complete cycle of the working substance, the net enternal energy of change.

$$\Delta E_{int} = 0$$

the net heat transfer per cycle:-

$$Q = |Q_H| - |Q_L|$$

We know:-

$$\Delta E_{int} = Q - W$$

$$0 = Q - W$$

$$\therefore W = Q$$

$$\boxed{\therefore W = |Q_H| - |Q_L|}$$

This is the net work done by the Carnot engine during a cycle.

Entropy Changes:—

⇒ There are two isothermal processes in each cycle of a carnot engine.

(1) During the isothermal expansion the working Substance absorbs heat $|Q_H|$ at temperature (T_H)

So, The increase in entropy :—

$$\Delta S_H = \frac{+|Q_H|}{T_H}$$

(2) During the isothermal compression the working Substance releases heat $|Q_L|$ at constant temperature (T_L)

So, The Decrease in entropy :—

$$\Delta S_L = \frac{-|Q_L|}{T_L}$$

Now, The net entropy change per cycle:—

$$\Delta S = \Delta S_H + \Delta S_L$$

$$\boxed{\Delta S = \frac{+|Q_H|}{T_H} + \frac{-|Q_L|}{T_L}}$$

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We know:—

For a Complete Cycle $\Delta S = 0$

$$0 = \frac{+|Q_H|}{T_H} + \frac{-|Q_L|}{T_L}$$

$$\frac{|Q_H|}{T_H} = \frac{|Q_L|}{T_L}$$

$$\frac{|Q_H|}{|Q_L|} = \frac{T_H}{T_L}$$

$$\boxed{\frac{|Q_L|}{|Q_H|} = \frac{T_L}{T_H}}$$

Efficiency of a Carnot Engine:— (কর্মদক্ষতা:)

We know—

Thermal efficiency of any engine is defined as,

$$\epsilon = \frac{\text{energy we get}}{\text{energy we provide}}$$

$$\boxed{\epsilon = \frac{|W|}{|Q_H|}} \qquad [W = |Q_H| - |Q_L|]$$

$$= \frac{|Q_H| - |Q_L|}{|Q_H|}$$

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$$\boxed{\therefore \epsilon = 1 - \frac{|Q_L|}{|Q_H|}}$$ → for any engine.

$$\boxed{\varepsilon = 1 - \frac{T_L}{T_H}}$$ → for Carnot engine.

In this case $T_L < T_H$ Always.

Problem no: 23:—

⇒ A Carnot engine whose low-temperature reservoir is at 17°c has an efficiency of 40%. By How much should the temperature of the high-temperature reservoir increased to increase the efficiency to 50%.

⇒ Is given:—

$T_L = 17°c = 290K$

$\varepsilon_1 = 40\%$

$\varepsilon_2 = 50\%$

we know:—

$$\varepsilon_1 = 1 - \frac{T_L}{T_{H_1}}$$

$$T_{H_1} = \frac{T_L}{1-\epsilon_1}$$

$$= \frac{290}{1-40\%}$$

$$= 483.33K$$

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Again we know, —

$$\varepsilon_2 = 1 - \frac{T_L}{T_{H_2}}$$

$$\therefore T_{H_2} = \frac{T_L}{1-\varepsilon_2}$$

$$= \frac{290}{1-50\%}$$

$$= 580 \text{ K}$$

So the increased temperature of the high temperature reservoir,

$$\Delta T_H = T_{H_2} - T_{H_1}$$

$$= (580 - 483.33) \text{ K}$$

$$= 96.67 \text{ K}$$

Problem: 24: —

⇒ A Carnot engine absorbs 52kJ as heat and exhausts 36kJ as heat in each cycle. Calculate

(a) The engines efficiency

(b) The work done per cycle in kilojoules.

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⇒ In given:—

$|Q_H| = 52 \text{ kJ}$

$|Q_L| = 36 \text{ kJ}$

(a)

we know:—

$$\varepsilon = 1 - \frac{|Q_L|}{|Q_H|} \times 100\%$$

$$= \left(1 - \frac{36}{52}\right) \times 100\%$$

$$= 30.76\%$$

(b)

we know:—

$$W = |Q_H| - |Q_L|$$

$$= (52 - 36) \text{ kJ}$$

$$= 16 \text{ kJ}$$

#Problem for Practice:—

Problem 25:—

⇒ A Carnot engine has an efficiency of 22%. It operatates between constant ~~temperature~~ temperature reservoirs differing in temperature by 75°C. What is the temperature of the

(a) lower - temperature

(b) Higher - temperature.

⇒ Is given:—

$$\Delta T = T_H - T_L = 75K$$
$$\varepsilon = 22\%$$

We know:— (b)

$$\varepsilon = 1 - \frac{T_L}{T_H}$$
$$= \frac{T_H - T_L}{T_H}$$
$$\therefore T_H = \frac{T_H - T_L}{\varepsilon}$$
$$= \frac{75}{22\%}$$
$$= 340.90K$$

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(a)

$T_H - T_L = 75K$

$\therefore T_L = T_H - 75$

$= 340.70 - 75$

$= 265.70 K$

Problem: 27:—

⇒ A Carnot engine operates between 235°C to 115°C absorbing 6.30×10^4 J per cycle at the higher temperature

(a) What is the efficiency of the engine?

(b) How much work per cycle is this engine capable for performing?

⇒ Is given:—

$T_H = 235 + 273K = 508K$

$T_L = 115 + 273 = 388K$

(a)

We know:—

$\varepsilon = \left(1 - \frac{T_L}{T_H}\right) \times 100\%$

$= \left(1 - \frac{388}{508}\right) \times 100\%$

$= 23.62\%$

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(b)

is given:—

$Q_H = 6.30 \times 10^4 J$

$\varepsilon = 23.62\%$

We know:—

$\varepsilon = \frac{W}{|Q_H|}$

$\therefore W = |Q_H| \times \varepsilon$

$= 23.62\% \times 6.30 \times 10^4$

$= 14.88 \times 10^3 J$

Ans:—

No-Perfect Engines:—

⇒ A perfect engines in only a dream; with Impossible requirements

$\varepsilon = \left(1 - \frac{T_L}{T_H}\right) \times 100\%$

$= \left(1 - \frac{0}{\infty}\right) \times 100\%$

$= 1 \times 100\%$

$= 100\%$